Nowcasting Using High Frequency Covariates

Ashley Hirashima, James Jones, Carl Bonham, Peter Fuleky

Department of Economics and UHERO
University of Hawaii at Manoa

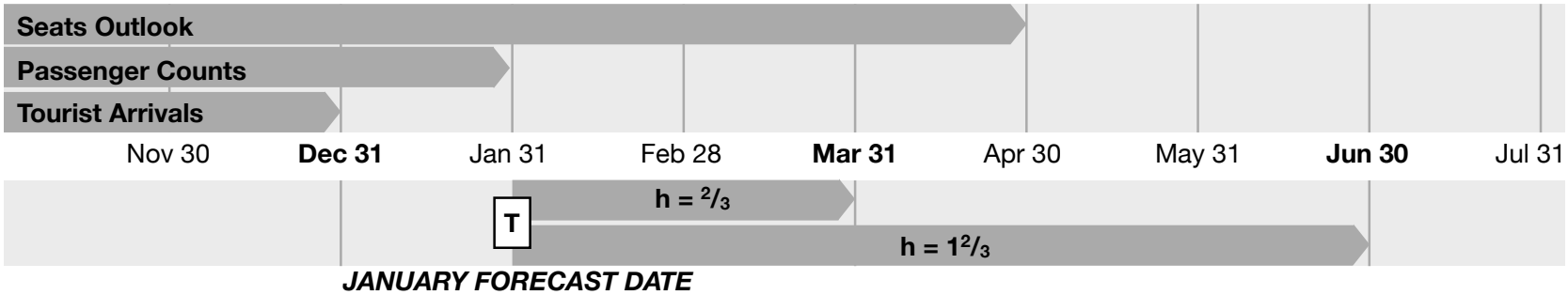
Bank of Thailand
December 15, 2017

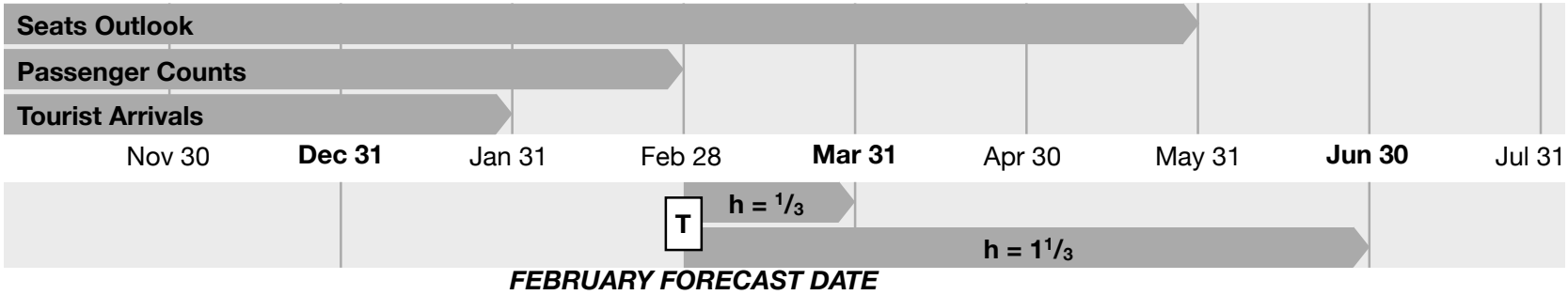
Motivation

Forecasting the Hawaii economy at UHERO.

- Macro models are generally specified at the quarterly frequency due to available data on state income and more recently GDP.
- Make use of a wide variety of indicators—sampled at different frequencies and available with varying lags.
- Aggregating indicators to the same frequency before estimation leads to loss of information.
- What is the best way to incorporate high frequency information?

Ragged Edge


Seats Outlook
Passenger Counts
Tourist Arrivals
Nov 30
Dec 31
Jan 31
Feb 28
Mar 31
Apr 30
May 31
Jun 30
Jul 31
T
h = 2/3
h = 1 2/3
JANUARY FORECAST DATE


Seats Outlook
Passenger Counts
Tourist Arrivals
Nov 30
Dec 31
Jan 31
Feb 28
Mar 31
Apr 30
May 31
Jun 30
Jul 31
T
h = 1/3
h = 1 1/3
FEBRUARY FORECAST DATE

Single Frequency Autoregressive Distributed Lag Models

Many univariate and multivariate models are nested within a general Autoregressive Distributed Lag (ADL) model (Song and Li, 2008):

$$y_t = \alpha + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \beta_0 x_t + \cdots + \beta_q x_{t-q} + \epsilon_t,$$

or

$$y_t = \alpha + \phi(L) y_{t-1} + \beta(L) x_t + \epsilon_t, \quad \text{for } t = p+1, p+2, \ldots,$$

with lag polynomials defined as

$$\phi(L) = \sum_{i=0}^{p-1} \phi_{i+1} L^i \quad \text{and} \quad \beta(L) = \sum_{j=0}^{q} \beta_j L^j.$$

Capturing reporting delays for the predictor:

$$y_t = \alpha + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \beta_{\Delta_x} x_{t-\Delta_x} + \cdots + \beta_q x_{t-q} + \epsilon_t$$

Forecasting with Autoregressive Distributed Lag Models

Obtain a forecast for horizon h based on information available at the forecast date T by evaluating

$$\hat{y}_{T+h} = \hat{\alpha} + \hat{\phi}(L) y_{T-\Delta_y} + \hat{\beta}(L) x_{T-\Delta_x} \tag{1}$$

- Δ_y and Δ_x account for the release lag: at time T, observations for the most recent periods may not be available.
- The coefficients in equation (1) are previously estimated in a regression that maintains equivalent lags of the predictors *relative to the response variable*:

$$y_t = \alpha + \phi(L) y_{t-h-\Delta_y} + \beta(L) x_{t-h-\Delta_x} + \epsilon_t$$

Mixed Frequency Autoregressive Distributed Lag Models

Relate a quarterly variable of interest to monthly predictors.

- Set the unit of time to a quarter, so the quarterly variable, y, is indexed with $t = 1, 2, \ldots$.
- Monthly indicators, x, are indexed with $t = \frac{1}{3}, \frac{2}{3}, 1, 1\frac{1}{3}, \ldots$.
- Use the fractional lag operator $L^{1/3}$ on monthly indicators:
  - x_2: value of monthly indicator in the last month of quarter 2,
  - $L^{1/3}x_2 = x_{1\frac{2}{3}}$ value for the second month of quarter 2, ...
- The model becomes:

  $$y_t = \alpha + \phi(L)y_{t-1} + \beta(L^{1/3})x_t + \epsilon_t \ , \quad \text{for} \quad t = 1, 2 \ldots \ ,$$

  with lag polynomials

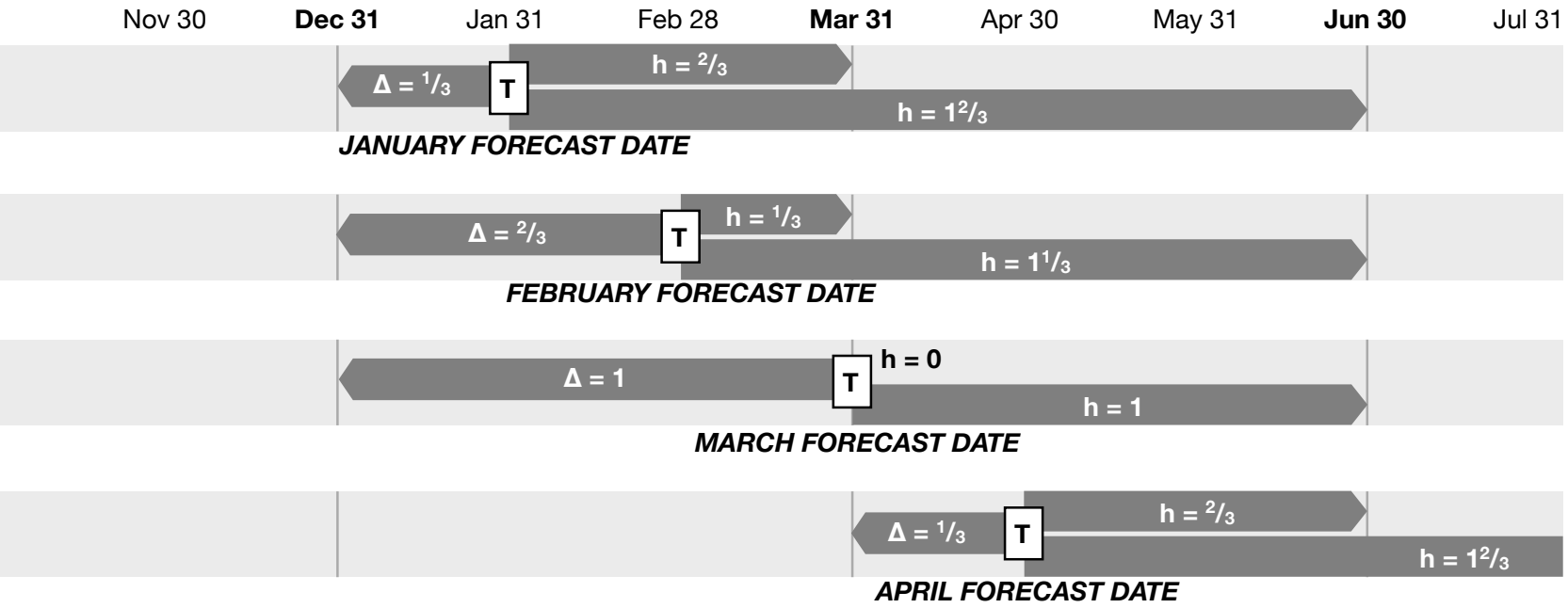
  $$\phi(L) = \sum_{i=0}^{p-1} \phi_{i+1} L^i \quad \text{and} \quad \beta(L^{1/3}) = \sum_{j=0}^{q} \beta_j L^{j/3}$$

Mixed Frequency Forecasts for Horizon h

At time T,

$$\hat{y}_{T+h} = \hat{\alpha} + \hat{\phi}(L) y_{T-\Delta_y} + \hat{\beta}(L^{1/3}) x_{T-\Delta_x} \ .$$

The forecast date T can fall at the end of any month. Therefore forecasts can be produced for horizons $h = \{0, \frac{1}{3}, \frac{2}{3}, 1, 1\frac{1}{3}, \ldots\}$.



Restrictions on UMIDAS models

Those above were Unrestricted MIxed DAta Sampling techniques (Foroni, 2015). We constrain the lag-polynomials $\phi(L)$ and $\beta(L^{1/3})$:

- *Autometrics* uses a wide variety of diagnostic tools to simplify a general unconstrained model (Hendry and Krolzig, 2004).
- *Non-overlapping predictors* are included based on their availability at time T.
- *MIDAS of Ghysels et al. (2007)* defines $\beta(L^{1/3})$ as an exponential Almon lag polynomial

$$\beta(L^{1/3}, \theta) = \beta_0(\theta)L^{0/3} + \beta_1(\theta)L^{1/3} + \ldots + \beta_q(\theta)L^{q/3},$$

where

$$\beta_j(\theta) = \frac{e^{\theta_1 j + \theta_2 j^2}}{\sum_{j=0}^{xmax} e^{\theta_1 j + \theta_2 j^2}} \ ,$$

so that only two hyper-parameters, θ_1 and θ_2, determine the distribution of weights along the lag polynomial.

Forecast Competition

- *Benchmark AR(4)* —a quarterly AR(4) model with no explanatory variables or monthly information.
- *Bridge equation*:
  - *Step 1*: an autoregressive model is used to iteratively forecast the values of the monthly regressors for the remainder of the quarter.
  - *Step 2*: a quarterly explanatory regression is estimated from historical data, and then evaluated with projected quarterly values of the regressors.
- *Monthly AR(p) or ADL* forecast aggregated to quarterly frequency. Only if the variable of interest is available at monthly frequency.

Data

- *Tourist arrivals:* monthly estimates are released with a one month lag. Quarterly arrivals are the sum of the monthly values.
- *Passenger counts* are available with a two-day lag. Include passengers arriving on both international and domestic flights but exclude flights originating in Canada. Monthly counts are sum of daily values.
- *Air seats outlook* captures the total number of scheduled seats (per month) expected for future direct flights to Hawaii. It is available for three months ahead.

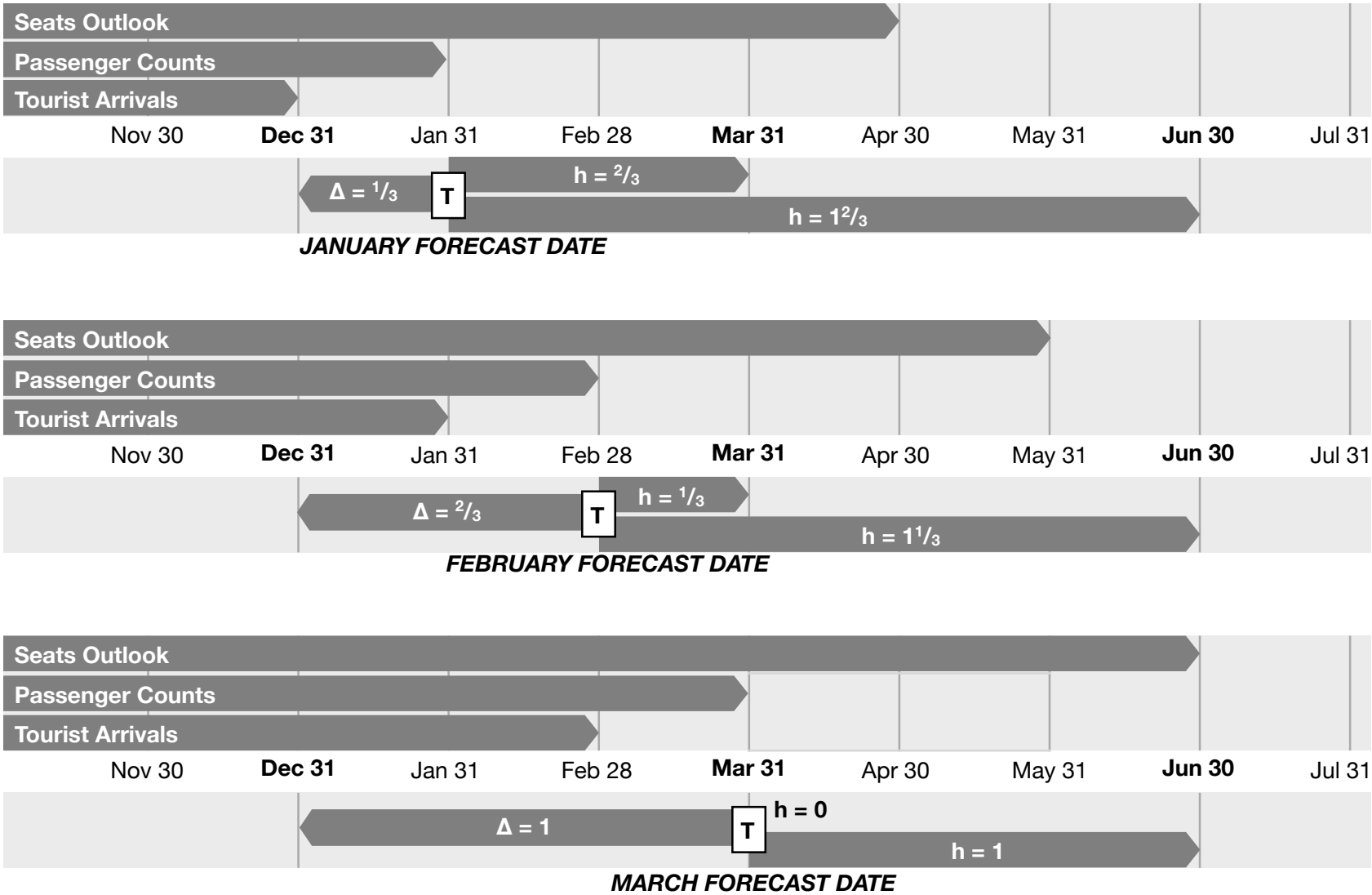
To avoid issues related to unit roots and seasonality, we applied year-over-year differences to log-levels of the series.

Estimation

- Our sample starts in January of 2001, and we produce quasi out-of-sample forecasts between January of 2008 and June of 2014.
- We estimate the model parameters from recursive samples where the starting period is held fixed and the ending period advances with the forecast date.
- The Autometrics based model, determined by diagnostic criteria, is respecified in each iteration of the forecasting exercise.
- The maximum lag length is 4 for quarterly tourist arrivals, and 12 for monthly tourist arrivals, passenger counts, and airline seats.
- The MIDAS model also uses these lag limits.

Information Set

As the forecast date, T, advances, the information set grows and the forecast horizon, h, shrinks.



Results

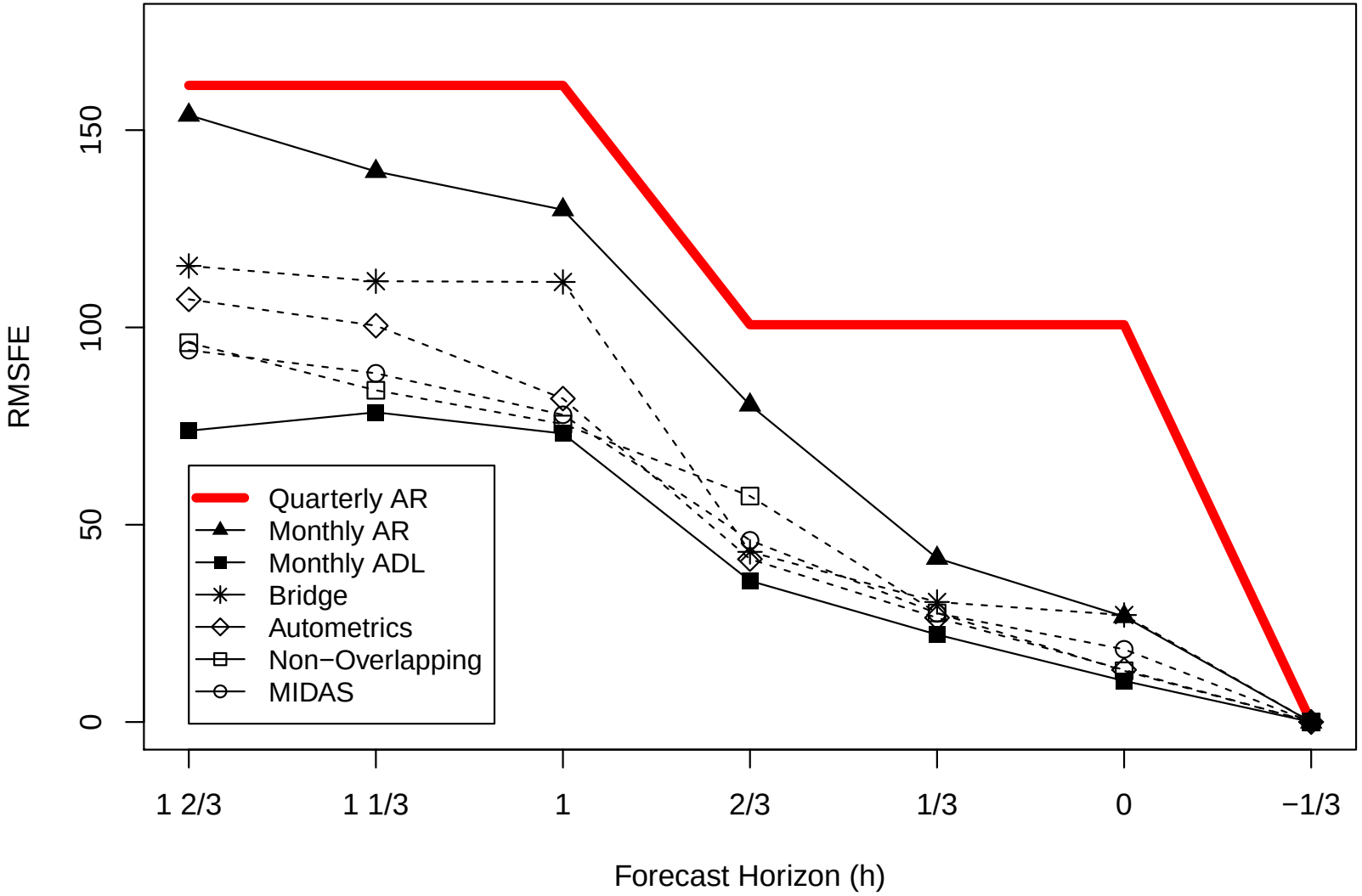


Figure: RMSFE for each model for forecast horizons between $h = 1\frac{2}{3}$ and $h = 0$.

Conclusions

- We find that all mixed frequency models, nowcasts and forecasts, are an improvement over the quarterly AR(4) model.
- Inclusion of the high frequency data does improve forecasts, which is in line with current research.
- No dominant method that performs the best overall when comparing mixed frequency models.
- The choice of model appears to be relatively unimportant, whereas the inclusion of the high frequency information is the main contribution to forecasting performance improvements.
- For practitioners, this conclusion implies that perhaps the best model is the simplest model that incorporates high frequency information.